

# Rocket Traj. Sim. Step 9

- If rocket is initially launched vertically, i.e. along position vector, then gravity turn must be initiated using a rotation. E.g. a gravity turn trajectory in x-y plane can be initiated using off-axial thrust modeled w/ rotation about the z-axis:

$$\vec{a}_{\text{thrust}} = \frac{T}{m_0 - \dot{m}_{\text{prop}} \cdot t} C_z \hat{r}$$

where

$$C_z = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Where

$$\hat{r} = [r_x, r_y, r_z] / \sqrt{r_x^2 + r_y^2 + r_z^2}$$

Python matrix:

```
import numpy as np
A = np.array([ np.cos(theta), -np.sin(theta), 0],
              [ np.sin(theta),  np.cos(theta), 0],
              [ 0,              0,          1])
```

Python vector:

```
B = np.array([rx],
              [ry],
              [rz])
```

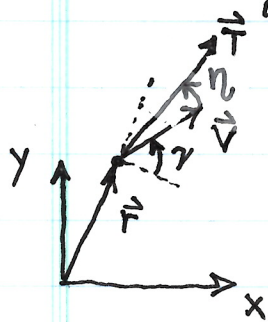
Python matrix multiplication:

```
C = np.matmul(A, B)
```

# Computing Steering Angle $\eta$ :

Step 9

path angle:  $\gamma = \frac{\pi}{2} - \arccos\left(\frac{\vec{r} \cdot \vec{v}}{|\vec{r}| |\vec{v}|}\right)$



E.g. 1: if  $\vec{r} = [1, 1]$  &  $\vec{v} = [2, 1]$

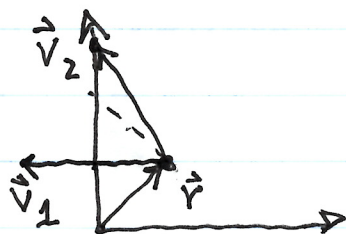
$$\gamma = \frac{\pi}{2} - \arccos\left(\frac{3}{\sqrt{2} * \sqrt{5}}\right)$$

$$= 71.6^\circ$$

E.g. 2: if  $\vec{r} = [1, 1]$  &  $\vec{v}_1 = [-2, 1]$

$$\gamma = \frac{\pi}{2} - \arccos\left(\frac{-1}{\sqrt{2} * \sqrt{5}}\right)$$

$$= -18.4^\circ$$



E.g. 3: if  $\vec{r} = [1, 1]$  &  $\vec{v}_2 = [0, 3]$

$$\gamma = \frac{\pi}{2} - \arccos\left(\frac{3}{\sqrt{2} * 3}\right)$$

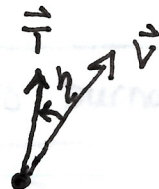
$$= 45^\circ$$

> 2D Traj. Sim. Initial Pitch Over:

$\eta > 0$  for launch az. =  $90^\circ$

$\eta < 0$  for " " =  $-90^\circ$

~~Applying  $\eta$  to Thrust Vector~~



Applying  $\eta$  to velocity norm vector, used in calculating thrust direction:

$$R = \begin{bmatrix} \cos \eta & -\sin \eta \\ \sin \eta & \cos \eta \end{bmatrix}$$

$$\hat{\vec{v}} = \begin{bmatrix} \hat{v}_x \\ \hat{v}_y \end{bmatrix} = \begin{bmatrix} v_x \\ v_y \end{bmatrix} / |\vec{v}| = \vec{v} / |\vec{v}|$$

$$\hat{\vec{v}}_\eta = R \hat{\vec{v}}$$

$$= \begin{bmatrix} \hat{v}_x \cos \eta - \hat{v}_y \sin \eta \\ \hat{v}_x \sin \eta + \hat{v}_y \cos \eta \end{bmatrix} = \begin{bmatrix} \hat{v}_{\eta x} \\ \hat{v}_{\eta y} \end{bmatrix}$$

Then,

$$\begin{aligned} \vec{a}_T &= \begin{bmatrix} a_x & a_y \end{bmatrix} \hat{\vec{v}}_\eta = a_x \hat{v}_{\eta x} + a_y \hat{v}_{\eta y} \\ &= \begin{bmatrix} a_x (\hat{v}_x \cos \eta - \hat{v}_y \sin \eta) \\ a_y (\hat{v}_x \sin \eta + \hat{v}_y \cos \eta) \end{bmatrix} \end{aligned}$$

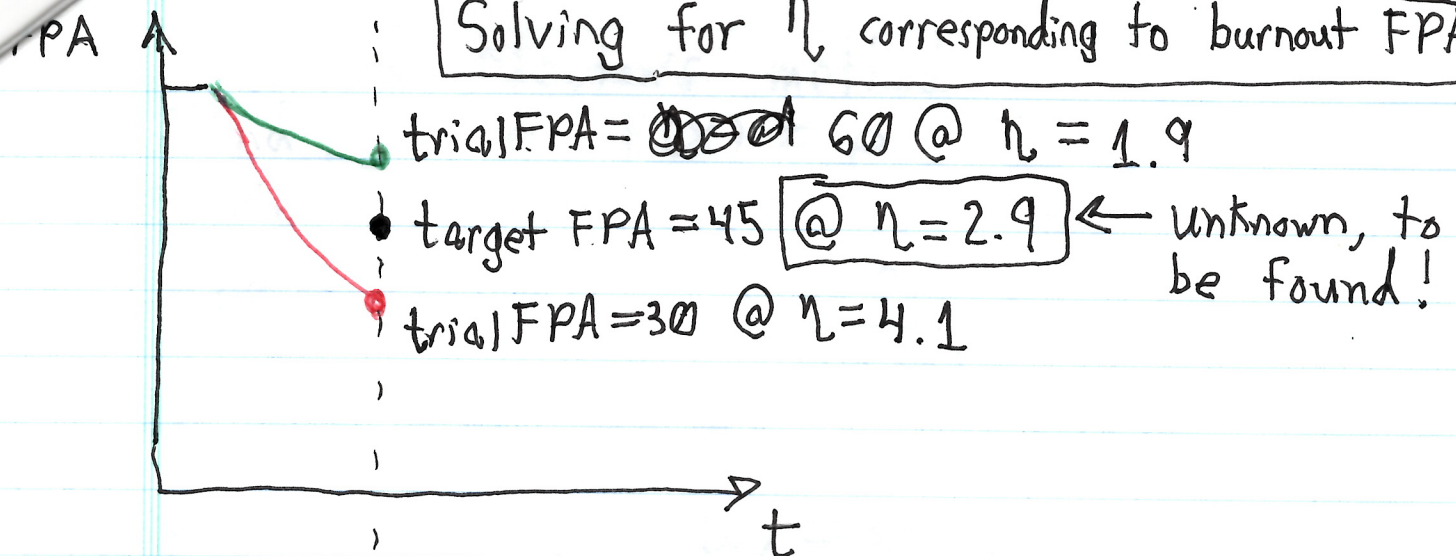
Dynamic Pressure:

$$q = \frac{1}{2} \rho v_{rel}^2$$



$$\alpha Z = 90^\circ \rightarrow \eta > 0$$

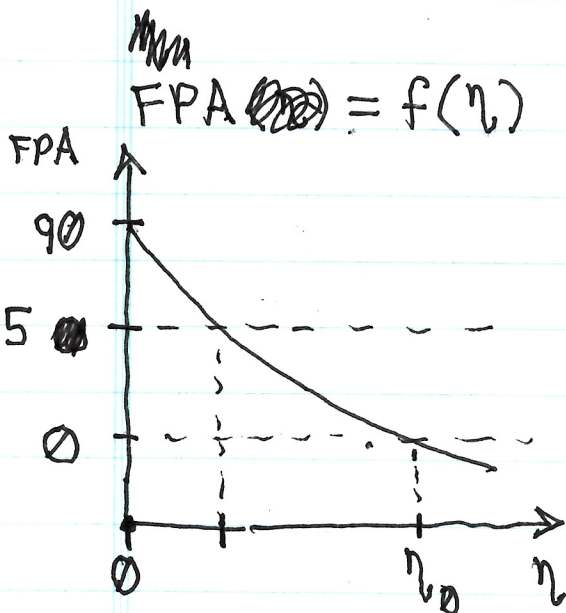
Solving for  $\eta$  corresponding to burnout FPA



~~FPA~~  $\eta_{\text{guess}} = 4.1$ :

$$\begin{aligned} \text{FPA diff} &= \text{trial FPA} - \text{target FPA} \\ &= 30 - 45 \\ &= -15 \end{aligned}$$

$$\begin{aligned} \eta_{\text{guess}} &= + \text{FPA diff} \times \frac{\eta_{\text{guess}} - \eta_{\text{target}}}{15} \\ &= + \text{FPA diff} \times \frac{1.2}{15} + \eta_{\text{guess}} \\ &\approx + \text{FPA diff} / 10 + \eta_{\text{guess}} \end{aligned}$$



⇒ Secant method for root finding

$$\begin{aligned} \eta_{n+1} &= \eta_n - \frac{f(\eta_n)}{f'(\eta_n)} \\ f'(\eta_n) &= \frac{f(\eta_n) - f(\eta_{n-1})}{\eta_n - \eta_{n-1}} \\ \eta_{n+1} &= \eta_n - \frac{f(\eta_n)(\eta_n - \eta_{n-1})}{f(\eta_n) - f(\eta_{n-1})} \end{aligned}$$

⇒ Modified secant method

$$\begin{aligned} \eta_{n+1} &= \eta_n + \frac{(p - f(\eta_n))(\eta_n - \eta_{n-1})}{f(\eta_n) - f(\eta_{n-1})} \\ \text{where } p &\text{ is target FPA} \end{aligned}$$

Modified Newton's method:

Find  $x$  where  $f(x) = p$

$$f(x_n) + f'(x_n)(x_{n+1} - x_n) = p$$

$$\cancel{f(x_n)} \quad x_{n+1} - x_n = \frac{p - f(x_n)}{f'(x_n)}$$

$$x_{n+1} = \frac{p - f(x_n)}{f'(x_n)} + x_n$$

E.g. Find  $x$  where  $f(x) = x^2 - 2 = 3$

$$p = 3, \quad x_0 = 1, \quad f(x_0) = -1, \quad f'(x_0) = 2$$

$$x_{\cancel{0}1} = \frac{3 + 1}{2} + 1 = 3$$

$$f(x_{\cancel{0}1}) = 7, \quad f'(x_{\cancel{0}1}) = 6$$

$$x_2 = \frac{3 - 7}{6} + 3 = 2.333$$

$$\cancel{f(x_2) = 47, \quad f'(x_2) = 94}$$
$$x_3 = \frac{3 - 47}{94} + 3 = 6.532$$

Analytical soln:

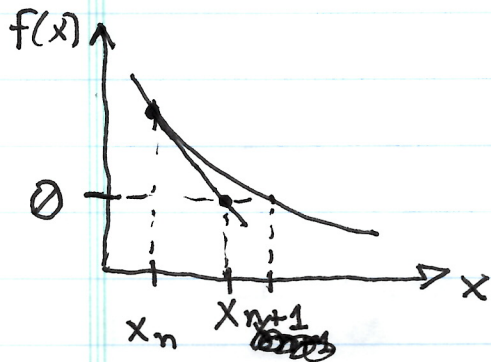
$$\cancel{x^2} = 5$$

$$x = \sqrt{5} = 2.24$$

Newton's method for root finding:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f'(x_n)(x_{n+1} - x_n) = -f(x_n)$$



$$f(x_n) + f'(x_n)(x_{n+1} - x_n) = 0$$

$$f(x) = \cancel{2x^2} \quad x^2 - 2$$

$$f'(x) = \cancel{4x} \quad 2x$$

Find root of ~~ln(x)~~  $f(x)$ :

$$x_0 = 0, f(x_0) = -2, f'(x_0) = 0$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0 - \frac{-2}{0}$$

$$x_0 = 1, f(x_0) = -1, f'(x_0) = 2$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1 - \frac{-1}{2} = 1.5$$

$$f(x_1) = 0.25, f'(x_1) = 3$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 1.5 - \frac{0.25}{3} = 1.42$$

Find where

Analytical soln:

$$2 = x^2$$

$$x = \sqrt{2}$$

$$= 1.41$$